

MOCK TEST: PHYSIOLOGY AND INJURIES IN SPORTS

PE Class 12 | Chapter 7 | Board Standards

TOTAL MARKS: 50 TIME: 1 HOUR SUBJECT CODE: 048

SECTION A: Very Short Answer (1m)

Q1. What is the primary function of the aerobic energy system during prolonged exercise?

1 Mark

Q2. Name one type of acute injury commonly seen in contact sports.

1 Mark

Q3. Define "delayed onset muscle soreness" (DOMS) in sports physiology.

1 Mark

Q4. List one physiological change in the cardiovascular system during exercise.

1 Mark

Q5. What is the role of PRICE in managing sports injuries?

1 Mark

SECTION B: Short Answer (3m)

Q6. Explain the anaerobic energy system and its importance in short-duration sports.

3 Marks

Q7. Discuss the causes and symptoms of a sprain. Mention one preventive measure.

3 Marks

Q8. Describe how the respiratory system adapts to regular exercise. Give one benefit.

3 Marks

Q9. How does the body maintain homeostasis during intense physical activity?

3 Marks

Q10. Outline the stages of injury rehabilitation. Mention one technique each.

3 Marks

SECTION C: Long Answer (5m)

Q11. Elaborate on physiological responses to exercise across Muscular, CV, and Respiratory systems.

5 Marks

Q12. Analyze types of sports injuries (acute vs chronic). Discuss prevention and care.

5 Marks

Q13. Explain the concept of recovery. Discuss active recovery, cryotherapy, and overtraining prevention.

5 Marks

ANSWER KEY / SOLUTIONS

A1: Breaks down carbs and fats using oxygen to provide sustained energy for long duration.

A2: Sprain or Fracture.

A3: Stiffness/pain felt 24-72 hours post-exercise due to micro-damage in muscle fibers.

A4: Increased stroke volume and heart rate to deliver more oxygenated blood.

A5: Protection, Rest, Ice, Compression, Elevation — used to control swelling/pain.

A6-10 Summary: Anaerobic fuels sprints (100m). Respiratory systems adapt by increasing vital capacity. Homeostasis is kept via sweating (thermoregulation). Rehab stages: Acute, Repair, and Remodeling.

A11-13 Insights: Chronic injuries like tennis elbow result from overuse. Prevention involves correct ergonomics and rest. Active recovery (light movement) clears lactic acid faster than total rest. Cryotherapy aids in reducing post-workout inflammation.

Mock Test Chapter 7 By Mindmaster Physical Education